Conversion of 3-SAT Clauses to Implication Graphs via Macro-Variable Substitution

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1 Conversion of Clauses to Implications with AND Gates

A 3-variable clause $(A \vee B \vee C)$ can be broken down into equivalent implication statements using only AND gates by applying De Morgan's laws:

- **Group 1 (Implications with AND gates as inputs):**
  - $(\neg A \wedge \neg B) \rightarrow C$
  - $(\neg A \wedge \neg C) \rightarrow B$
  - $(\neg B \wedge \neg A) \rightarrow C$
  - $(\neg B \wedge \neg C) \rightarrow A$
  - $(\neg C \wedge \neg A) \rightarrow B$
  - $(\neg C \wedge \neg B) \rightarrow A$
- **Group 2 (Implications with OR gates converted to AND gates):**
  - $\neg C \rightarrow \neg(\neg A \wedge \neg B)$
  - $\neg A \rightarrow \neg(\neg B \wedge \neg C)$
  - $\neg B \rightarrow \neg(\neg A \wedge \neg C)$

2 2-Variable Substitution Assumption

Each distinct AND gate block is assumed to be a single substitute variable (referred to as a macro-switch):

- Let $X = (\neg A \wedge \neg B)$
- Let $Y = (\neg B \wedge \neg C)$
- Let $Z = (\neg A \wedge \neg C)$

By substituting these macro-variables, the complex 3-variable rules are compacted into structural placeholders fit for directed graph representation.

3 Running Tarjan's Algorithm on the Implication Graph

We construct a directed graph $G = (V, E)$ where the vertices V represent the literal variables and the macro-switches (A, B, C, X, Y, Z and their respective negations), and the edges E represent the implication arrows. Tarjan's algorithm is run on this directed graph in linear time $\mathcal{O}(V + E)$ to isolate all Strongly Connected Components (SCCs).

4 Macro-Variable Expansion and Pre-Assignment Substitution

Once Tarjan's algorithm groups the vertices into initial strongly connected components, the macro-switches must be fully resolved. Before performing any contradiction checks or evaluating assignments, every macro-switch located inside an SCC is structurally expanded into its underlying micro-variable definition.

For example, if a macro-switch is defined as:

$$X = (\neg A \wedge B)$$

Substituting the definition of X back into its component group flattens the SCC, populating it directly with its constituent micro-variables:

$$\ldots, \neg A, B, \ldots$$

This pre-assignment expansion strips away the structural placeholders so that the core micro-variables govern the final assessment.

5 Contradiction Checking and 2-SAT Value Assignment

Following the flattening of all components, the graph's satisfiability is determined by evaluating the micro-variables in the exact manner of a standard 2-SAT workflow:

1. **Flattened Contradiction Check:** The expanded components are checked for explicit logic violations. If any component contains both a base micro-variable and its immediate literal complement simultaneously (e.g., both A and $\neg A$ wind up in the same group due to macro-expansion), a direct logical contradiction is triggered. The system flags this state as **UNSAT**.
2. **Reverse Topological Assignment:** If no contradictions are found, a satisfying assignment is guaranteed. The components are traversed in **reverse topological order** (from sinks to sources, matching the natural finish-time output of Tarjan's algorithm):

- The micro-variables inside the current unassigned component are greedily assigned a truth value of **True**.
- Their complementary literals residing in the conjugate component are automatically assigned **False**.

Because the macro-switches are entirely expanded into their micro-variables prior to this phase, the resulting truth assignments inherently satisfy all underlying 3-SAT clause dependencies without requiring secondary macro-validation.